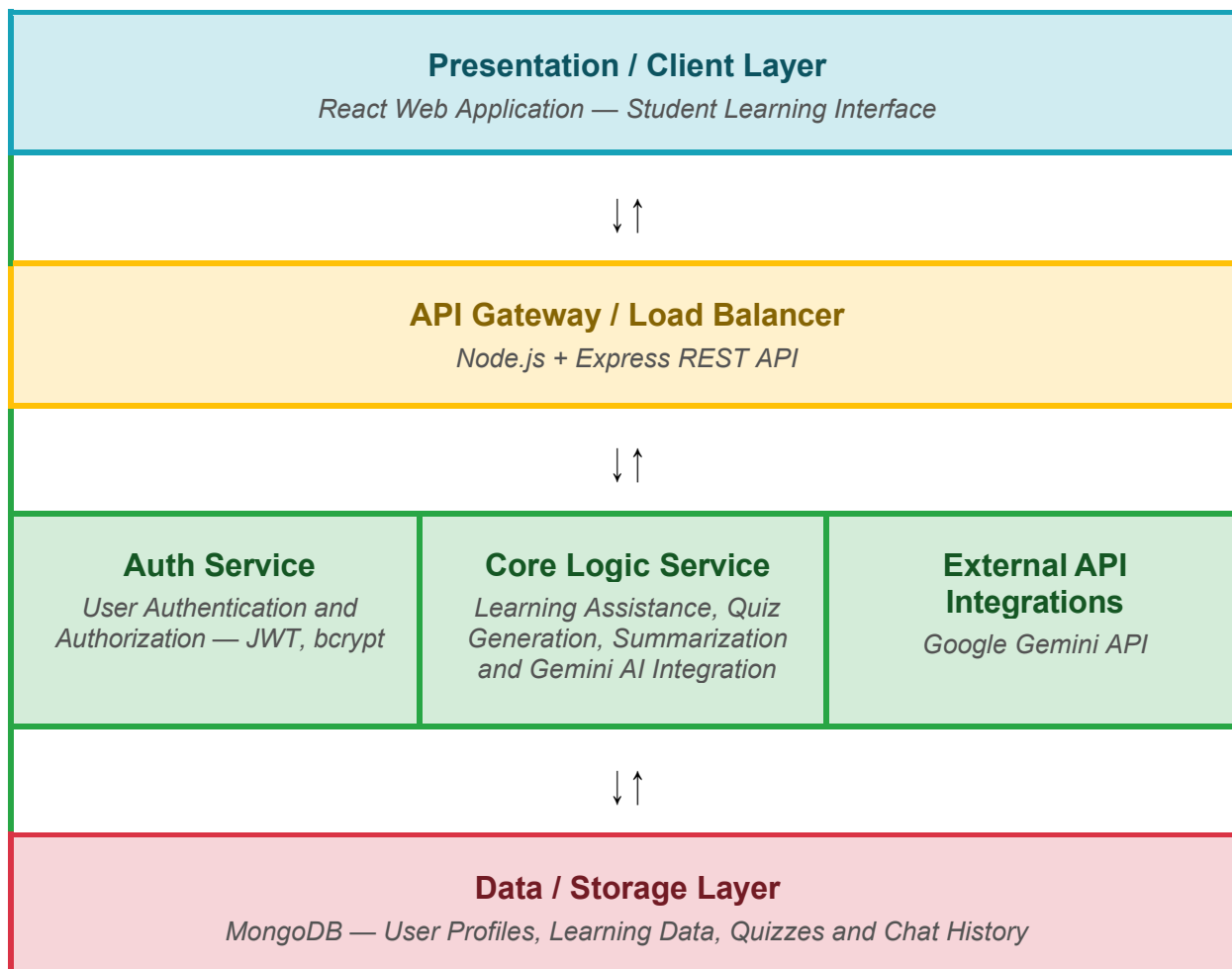


## Solution Architecture

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 30 September 2026                                   |
| <b>Team ID</b>       | SWTID-2026-8336                                     |
| <b>Project Name</b>  | EduGenie — Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                             |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name            | Description / Role in Architecture                                                                                                                          | Technologies Used                                          |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <i>Presentation Layer</i> | <i>Provides the web interface for students to access learning assistance, ask questions, generate quizzes, summarize content and view learning results.</i> | <i>React.js, HTML, CSS, JavaScript</i>                     |
| <i>API Gateway</i>        | <i>Receives requests from the frontend, manages API communication and routes requests to the appropriate backend services.</i>                              | <i>Node.js, Express.js, REST API</i>                       |
| <i>Microservices</i>      | <i>Handles authentication, learning assistance, quiz generation, content summarization and communication with the Google Gemini API.</i>                    | <i>Node.js, Express.js, JWT, bcrypt, Google Gemini API</i> |
| <i>Database</i>           | <i>Stores user information, learning activities, generated quizzes, learning content and chat history for personalized learning support.</i>                | <i>[MongoDB</i>                                            |